

EY Physical AI Engineering | Independent candidate materials

Russell Dudek

Physical AI Reality Transfer Review

Pittsburgh, Pennsylvania | 412.287.8640 | russelldudek@gmail.com linkedin.com/in/russelldudek | <https://russelldudek.github.io/EY/>

Role-specific review artifact

Does the capability earn the right to act?

A review canvas for connecting simulation, embodiment and client operation.

A

Use

Complete this review at meaningful workstream transitions. It is intentionally evidence-light: link to the actual architecture, test, telemetry, risk and operating artifacts rather than copying them into a new system.

Evidence dimension

Required question

Evidence link / owner

State

Mission Outcome and task

What job is the system being allowed to perform, for whom, under which operating conditions?

_____ Owner: _____

Unknown / framed / accepted

Environment Twin and coverage

Which states, variants, disturbances and edge conditions are represented or deliberately excluded?

_____ Owner: _____

Gap / rehearsal / covered

Perception Knowledge and uncertainty

What can the system observe, infer, misclassify or fail to see? How is degradation detected?

_____ Owner: _____

Opaque / instrumented / bounded

Authority Control and safety

Who may initiate, override, stop, resume and accept residual risk?

_____ Owner: _____

Conflict / explicit / rehearsed

Run state Release and recovery

Can behavior be traced to model, code, configuration and environment? Can the system roll back and recover?

_____ Owner: _____

Fragile / observable / recoverable

Learning Adoption and economics

Who owns support, incidents, workflow adoption, value evidence and the next improvement?

_____ Owner: _____

Unowned / assigned / sustained

Current disposition

■ Rehearse ■ Instrument ■ Bound ■ Pilot ■ Hold ■ Operate ■ Reuse

Next evidence

Decision owner: _____ Evidence due: _____ Review date: _____

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Scenario and consequence guide

Review the weakest evidence first

The goal is not a composite score. The weakest consequential dimension should shape the next workstream decision.

B

Observed condition

Likely weak dimension

Recommended disposition

Evidence to seek

Behavior succeeds in the twin but fails around a novel obstacle.

Environment / perception

Rehearse

Scenario expansion, uncertainty behavior, controlled physical test.

Sensor confidence degrades gradually without a visible alarm.

Perception / run state

Instrument

Health telemetry, drift detection, diagnostic and fallback behavior.

Operator override is possible but responsibility for resuming autonomy is unclear.

Authority

Bound

Control modes, role definition, stop/resume protocol and rehearsal.

A release changes motion behavior despite passing unit tests.

Run state

Hold

SIL/HIL regression, configuration trace, rollback and recovery proof.

Technical performance is strong but users bypass the workflow.

Mission / learning

Reframe or pilot

Workflow observation, adoption friction, human role and value baseline.

One client deployment works but support and economics are unclear.

Learning

Stabilize before reuse

Owner, support model, recurring effort, incident pattern and reusable asset boundary.

Workstream decision brief

What changed? _____

Why does it matter in physical operation? _____

What does the current pattern protect? _____

What evidence is sufficient for the next bounded step?

What remains under human authority? _____

What gets learned or reused? _____

Integrity rule

Do not average away a safety, authority or run-state failure. Do not use an invented maturity score.
Record the evidence, the unresolved assumption, the accountable owner and the next bounded decision.

Independent candidate artifact. Illustrative structure only; it does not represent EY methodology or client facts.